Demonstrating Conduction

Exam code: 0625 & 0972

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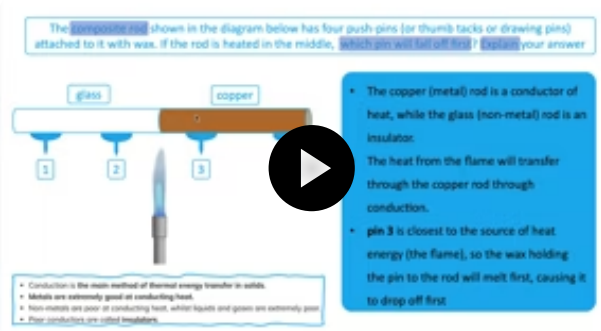
Updated on 28 March 2025



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Experiments demonstrating thermal conductors

Good and bad thermal conductors

- Good thermal conductors are solids which easily transfer heat
  - For example; an aluminium pan or copper wire
- Bad thermal conductors (also called insulators) are solids which do not transfer heat well
  - For example; a wool blanket or layers of cardboard or paper

Comparing conduction in tiles and textiles

- This conduction experiment shows why homes use rugs and carpets
- Find a tiled or stone area of floor
  - In the same room leave a rug or bath towel (not a thin cloth, it must be thick)
  - The textile must stay in place on the floor for several hours to ensure they are at thermal equilibrium (the same temperature)

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it
Help me understand

- Observe the apparent temperature of the two materials as felt through the feet
- It will feel as though the tiles are cold while the rug is warm, however, they are at exactly the same temperature

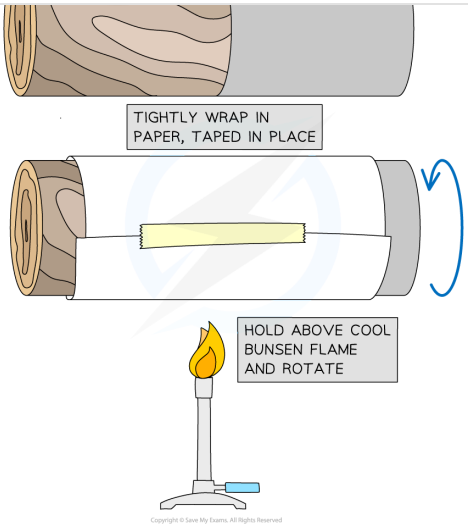
Energy is transferred by heating from the hotter foot to the cooler tiles by conduction

Explanation

- Tiles and stone are good conductors of heat
  - Where the foot touches the tiles, **heat is transferred away** from the foot, making it **feel** cold
  - The **foot** has become colder since it lost heat to the tiles
- Textiles such as rugs are good insulators, meaning they are poor conductors of heat
  - Where the foot touches the rug, heat is not transferred away from the foot
  - This foot feels relatively warmer than the one which has lost heat to the tiles
  - The foot has stayed at its starting temperature

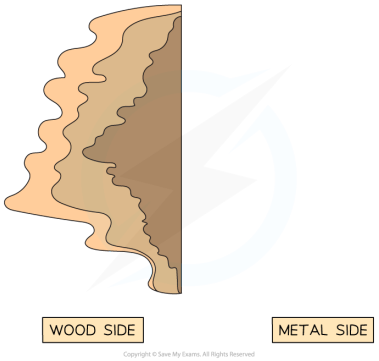
Comparing conduction in wood and paper

- A cylindrical rod made of half wood and half metal is wrapped tightly in paper



Method for showing different thermal conduction of wood and metal

- Using a gentle flame, and holding the rod clear of the top of the flame, gently heat the paper at the join of the wood and metal
  - Turn the rod so that the paper is well-heated all around the circumference of the rod
  - Stop when the paper is clearly discoloured
- Remove the rod from the flame, gently unwrap the paper and observe the burn pattern
  - A distinct pattern is seen;
    - Where the paper touched the metal surface it is undamaged
    - Where the paper touched the wood surface it is charred



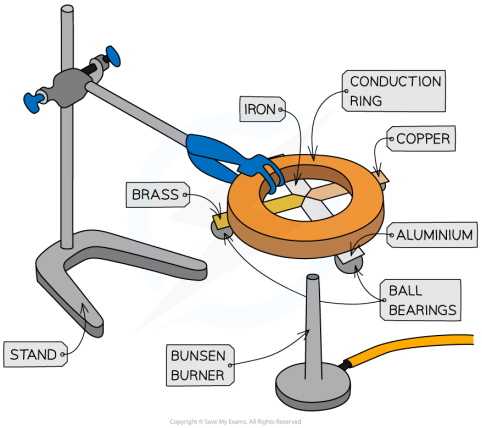
Explanation

- Metal is a good conductor of heat
  - Where the paper touched the metal in this conduction experiment, heat was transferred from the paper into the metal and along the length of the metal
  - This prevented the paper getting hot

transferred from the paper
    - This meant that the paper did get hot enough to start to burn

Demonstrating different rates of thermal conduction in metals

- A simple conduction experiment to demonstrate the relative conducting properties of different materials can be carried out using apparatus similar to that shown in the diagram below



The above apparatus consists of 4 different metal strips of equal width and length arrange around an insulated circle, holding ball bearings with wax

- Ball bearings can be stuck to each of the strips and equal distance from the centre, using a small amount of wax
- The strips should then be turned upside down and the centre heated gently using a candle, so that each of the strips is heated at the point where they meet
- When the heat is conducted along to the ball bearing, the wax will melt and the ball bearing will drop
- By timing how long this takes for each of the strips, their relative thermal conductivities can be determined

Examiner Tips and Tricks

With two of the demonstrations above (the rug & tiles and the rod made of wood and metal) students often get confused, as the result may 'feel' as though it is the wrong way round. Think about the movement or transfer of the heat.

The more heat is taken away, the cooler something will be. That is why the paper doesn't burn when it is next to the metal (although usually everyone guesses that it will!)

The more heat is held in by a layer of insulation, the more heat remains, which is why rugs and carpets help your feet to feel warm.

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